

Sabrina Marbut Appel

NSF Astronomy & Astrophysics Postdoctoral Fellow
Department of Astrophysics – American Museum of Natural History
200 Central Park West, New York, NY 10024
sappel@amnh.org – sabrinaappel.github.io

APPOINTMENTS

2024 - Present **NSF Astronomy & Astrophysics Postdoctoral Fellow**
American Museum of Natural History (AMNH), New York, NY

EDUCATION

2018 - 2024 **Rutgers, The State University of New Jersey**, Piscataway, NJ
Ph.D. in Physics and Astronomy, 2024
M.S. in Physics and Astronomy, 2022
Thesis Title: “Exploring the Effect of Protostellar Feedback on
Star Formation and Molecular Cloud Dynamics”
Thesis Defense: Jun. 20, 2024
Thesis Advisor: Dr. Blakesley Burkhart

2013 - 2017 **Reed College**, Portland, OR
B.A. in Physics, 2017
Senior Thesis: “Simulating the Gravitational Lensing of Massive
Particles: An Exploration of Scattering Solutions of the
Schwarzschild Metric”
Senior Thesis Advisor: Dr. Andrew Larkoski

2009 - 2013 **South Salem High School**, Salem, OR
International Baccalaureate Diploma, 2013

OTHER RESEARCH EXPERIENCE

Summer 2016 **American Museum of Natural History**
NSF Funded REU Participant, Advised by Dr. Dave Zurek

Summer 2015 **Rutgers, The State University of New Jersey**
NSF Funded REU Participant, Advised by Dr. Andrew Baker

GRANTS

2024 - 2027 PI for the **NSF AAPF** Proposal “Exploring the Impact of Protostellar
Jets in Star Cluster Formation and Evolution”, Awarded for Sep. 1,
2024 to Aug. 31, 2027 and hosted at AMNH

2023 Collaborator on the **NSF Proposal** “The Untimely Deaths of Star
Clusters”, submitted Fall 2022, awarded Fall 2023

SCHOLARSHIPS, HONORS, AND AWARDS

Fall 2023 **Noemie Koller Scholarship**, Rutgers Physics and Astronomy De-
partment

Summer 2023 **SGS Travel Award**, Rutgers School of Graduate Studies, awarded
for participation in the “Olympian Symposium”

Summer 2022 **Torrey Fellowship**, Rutgers Physics and Astronomy Department

Spring 2022	Peter Lindenfeld Graduate Fellowship , Rutgers Physics and Astronomy Department
Summer 2019	Boyd Scholarship , Rutgers Physics and Astronomy Department
Summer 2019	Travel Award , Rutgers Physics and Astronomy Department, awarded for travel to the “Advancing Theoretical Astrophysics Summer School”
2017	Phi Beta Kappa , inducted to Reed College Chapter May 15, 2017
2014 - 2017	NSF Scholar , awarded by Reed College for the 2014/15, 2015/16, and 2016/17 academic years
2013 - 2017	President’s Commendation for Academic Excellence , Reed College, awarded for academic performance in the 2013/14, 2014/15, 2015/16 and 2016/17 academic years
2013 - 2017	Reed College Grant , Reed College Financial Aid Office, awarded for the 2013/14, 2014/15, 2015/16, and 2016/17 academic years

STUDENT MENTORING

2026 to present	Donglin Wu , Independently-funded Summer Researcher (Summer 2026) & Ongoing collaboration, AMNH, New York, NY
2026 to present	Edwin Lliguicota , AstroCom NYC Participant (Summer 2026) & Ongoing collaboration, AMNH, New York, NY
2025 to present	Fred Javier , AMNH Physical Sciences REU Participant (Summer 2025), Independently-funded Summer Researcher (Summer 2026), & Ongoing collaboration, AMNH, New York, NY
2021 - 2022	Avery Kiihne , Undergraduate Researcher, Summer 2021 - Summer 2022, Rutgers University, Piscataway, NJ

SERVICE

2026	NSF Review Panel , Participant, Astronomy Section
2025 to present	Seminar Committee , AMNH, New York, NY
2024 to present	Paper Referee Astronomy & Astrophysics (A&A)
2020 - 2024	Leadership Roles in MiPA (Minorities in Physics and Astronomy group), Rutgers University, Piscataway, NJ Past President - Sep. 2023 to Aug. 2024 President - Dec. 2022 to Sep. 2023 Graduate Chair - Summer 2020 to Dec. 2022 Co-Coordinator of EIJC (See below) - 2020 to 2022 - Led the creation of a MiPA Charter and Code of Conduct (2020)
2020	Founder of the Equity and Inclusion Journal Club (EIJC) , Rutgers University, Piscataway, NJ
2019 - 2020	Officer of Women in Physics and Astronomy (WiPA) , Rutgers University, Piscataway, NJ Co-led the establishment of weekly meetings and the name change to Minorities in Physics and Astronomy (in Spring/Summer 2020)
2019 - 2020	Co-President of the Physics and Astronomy Graduate Student Organization , Rutgers University, Piscataway, NJ

2020 **Participant in WFIRST Advocacy Day**, as a representative of Rutgers University, Washington D.C, (March 5, 2020)

CODE CONTRIBUTIONS & SOFTWARE SKILLS

- User and Developer of the Torch code since 2019; website: torch-sf.bitbucket.io
- Experience with `git`, `bitbucket`, & `github` since 2019
- Languages: `python` (~ 10 years); `Fortran 90` (~ 7 years); `html` (> 4 years); etc.
- Experience with high performance computing since 2018

OUTREACH AND TEACHING (SELECTED)

Jul. 30, 2026	Workshop Facilitator: “Code Like an Astrophysicist: Computational Thinking in your Science Classroom,” CTLE Professional Learning for K-12 Teachers, AMNH, New York, NY
May 1, 2026	Workshop Instructor: Command Line Workshop, AstroCom NYC Program, AMNH, New York, NY
April 18, 2026	Coffee with a “Curator” Classroom Visit: “Born Among the Stars: Exploring the Physics of Star Formation,” Lang Science Program, AMNH, New York, NY
Feb. 21, 2026	Classroom Visit: “Born Among the Stars: Exploring the Physics of Star Formation,” Lang Science Program: <i>Stars Course</i> , AMNH, New York, NY
Nov. 20, 2025	Guest Lecturer: <i>Astronomy 110</i> , Course Instructor: Dr. Saavik Ford, CUNY BMCC, New York, NY
Oct. 14, 2025	Public Lecture: “Born Among the Stars: Exploring the Physics of Star Formation,” Amateur Astronomers Association Lecture (Virtual), New York, NY
Oct. 21, 2024	Guest Presentation: “Using Computational Astrophysics to Understand Star Formation,” Visit to Computational Astrophysics Course including a talk, Rutgers University, New Brunswick, NJ
Fall 2022	Teaching Assistant , <i>Computational Astrophysics</i> , Rutgers University, Piscataway, NJ
Fall 2020	Teaching Assistant , <i>Byrne Seminar: The Rutgers Undergraduate Pipeline to Research & Education in Physics (RU-PREP)</i> , Rutgers University, Piscataway, NJ
	Includes being available as a mentor for the students
Spring 2019	Teaching Assistant , <i>Extended Analytical Physics II</i> , Rutgers University, Piscataway, NJ
Fall 2018	Teaching Assistant , <i>Extended Analytical Physics I</i> , Rutgers University, Piscataway, NJ
Fall 2017 - Summer 2018	Volunteer , Physics Lab & Featured Exhibits, Oregon Museum of Science and Industry, Portland, OR
Fall 2014 - Spring 2017	Teaching Assistant , <i>Introductory Physics Laboratory</i> , Reed College, Portland, OR

Fall 2014 - **Drop-in Tutor**, *Introductory Physics*, Reed College, Portland, OR
 Spring 2015
 Summer 2014 **Library Intern**, Salem Public Library's Children's Desk, Salem, OR
 Fall 2013 - Math and Reading **Tutor**, Grout Elementary School, Portland, OR
 Spring 2014

CONFERENCE ORGANIZATION

Nov. 21, 2025 Head of the SOC/LOC, **Physical Sciences Jamboree**, AMNH, New York, NY
 Oct. 20-24, 2025 SOC/LOC, **Torch Developers Meeting**, AMNH, New York, NY
 Apr. 22-26, 2024 LOC, **Advanced AMUSE Workshop**, AMNH, New York, NY
 Oct. 21, 2022 Head of the SOC/LOC, **Torch Regional Meeting**, CCA, Flatiron Institute, New York, NY
 Aug. 17-19, 2022 Head of the SOC/LOC, **Torch Workshop**, CCA, Flatiron Institute

TALKS (SELECTED)

Sept. 22, 2026 **Invited Interview:** AAS Journal Author Series, AAS Youtube Channel, <https://youtu.be/1GM2dcabosQ>
 Jan. 4, 2026 "Exploring Protostellar Jets in Star Cluster Formation with Torch," NSF AST Fellow Symposium, AAS 247 Winter Meeting, Phoenix, AZ
 Nov. 21, 2025 Intro. Slide, Physical Sciences Jamboree, AMNH, New York, NY
 Jul. 24, 2025 "The Problem of Star Formation: Star Particles, Sink Particles, and How to Make a Stellar Population," Caveat Talk, Numerical Recipes in Star Formation: Lorentz Center Workshop, Leiden, The Netherlands
 Jul. 17, 2025 "Exploring Protostellar Jets in Star Cluster Formation with Torch," Interstellar Institute 7: A three week scientific session at Institut Pascal, Paris-Saclay, France
 May 28, 2025 **Invited Talk:** "The Cycle of Star Formation across Scales: Stellar Feedback as a Source of Interstellar Turbulence," Star Formation, Stellar Feedback, and the Ecology of Galaxies, Visegrád, Hungary
 Jan. 15, 2025 "Exploring the Role of Protostellar Jets in Simulations of Star Cluster Formation using Torch," Oral Session of the AAS 245 Winter Meeting, National Harbor, MD
 Jan. 12, 2025 "Exploring the Impact of Protostellar Jets in Star Cluster Formation and Evolution," NSF AST Fellow Symposium, AAS 245 Winter Meeting, National Harbor, MD
 Dec. 12, 2024 "The Impact of Protostellar Jets in Star Cluster Formation Simulations," CICO-VICO-CCA Origins Workshop, Center for Computational Astrophysics, Flatiron Institute, New York, NY
 Sep. 17, 2024 **Invited Talk:** "Star Formation at the Cloud-Scale: Understanding the Density Distribution and the Gas Dynamics," KITP: Cosmic Origins: The First Billion Years, UC Santa Barbara, Santa Barbara, CA
 Jun. 20, 2024 "Exploring the Effect of Protostellar Feedback on Star Formation and Molecular Cloud Dynamics," Ph.D. Thesis Defense, Rutgers University, Piscataway, NJ

Mar. 21, 2024	“Exploring the Star Formation Rate and the Effect of Protostellar Feedback at the Cloud-scale,” Thursday Seminar, Columbia University, New York, NY
Jan. 11, 2024	“The Density Distribution and the Gas Compression and Expansion Rates of Molecular Clouds: Understanding the Star Formation Rate and Exploring the Effect of Protostellar Feedback,” Dissertation Talk at the 243rd AAS Meeting, New Orleans, LA
Jun. 28, 2023	“How the Gas Dynamics Set the Star Formation Rate of Molecular Clouds,” The 2023 Northeast Star and Planet Formation Meeting, Center for Astrophysics (CfA), Cambridge, MA
Jun. 2, 2023	“How the Gas Dynamics Set the Star Formation Rate of Molecular Clouds,” The Olympian Symposium, Paralia Katerini, Mt. Olympus, Greece
May 26, 2023	Invited Talk: “How the Gas Dynamics Set the Star Formation Rates of Molecular Clouds,” Young MMF talk at the Midwest Magnetic Fields Workshop 2023, Online
Apr. 20, 2023	Invited Talk: “How the Gas Dynamics Set the Star Formation Rates of Molecular Clouds,” Princeton Thunch Series, Princeton University, Princeton, NJ
Mar. 21, 2023	Invited Talk: “How the Gas Dynamics Set the Star Formation Rates of Molecular Clouds,” KITP: Conference on Galaxy Formation and Evolution in the Data Science Era, UCSB, Santa Barbara, CA
Aug. 23, 2022	“The Impact of Stellar Feedback on the Dynamics and Evolution of Star Forming Regions,” the Clusters Workshop at McMaster University, Hamilton, Ontario, Canada
Aug. 18, 2022	“Implementing Protostellar Outflows in TORCH,” the Summer 2022 Torch Workshop, Center Computational Astrophysics, Flatiron Institute, New York, NY
Jul. 21, 2022	“The Impact of Stellar Feedback on the Dynamics and Evolution of Star Forming Regions,” With Two Eyes: A three week scientific session of the Interstellar Institute, Institut Pascal, Paris-Saclay, France
Dec. 9, 2022	“The Impact of Stellar Feedback on the Density PDF in Star Forming Regions,” VICO-CICO Fall 2021 Workshop, University of Virginia, Charlottesville, VA
Dec. 3, 2021	“The Impact of Stellar Feedback on the Density PDF in Star Forming Regions,” The Mid-Atlantic Section of the APS Meeting, Rutgers University, Piscataway, NJ
2019 - 2021	“Women in Physics: A Case Study of Equity Issues in Physics,” Student Seminar in Physics and Astronomy, Rutgers University, Piscataway, NJ (joint talk with Charlotte Olsen) Gave the same talk Mar. 14, 2019, Nov. 21, 2019, Oct. 22, 2020 (Online), and Dec. 2, 2021 (Online)

- Jul. 26, 2021 “Towards an Analytic Model of Star Formation: What Makes Star Formation Inefficient?” Interstellar Institute’s program “The Grand Cascade”, Paris-Saclay University’s Institut Pascal, Online
- Dec. 14, 2020 “Towards an Analytic Model of Star Formation: What Makes Star Formation Inefficient?” CICO-VICO Fall 2020 Workshop, Online
- Sep. 6, 2019 “Investigating the Impact of Stellar Feedback on Models of Star Formation,” Gotham Fest 2019, Simons Foundation Center for Computational Astrophysics, New York, NY
- May 2, 2017 “Simulating Gravitational Lensing,” Senior Thesis Oral Examination, Reed College, Portland, OR
2-hour exam before an interdisciplinary, 4-person board of faculty

POSTERS AND OTHER PRESENTATIONS

- Apr. 15-19, 2024 “Exploring the Star Formation Rate and the Effect of Protostellar Feedback at the Cloud-scale,” Poster, **2024 Spring Symposium – Recipes to Regulate Star Formation at All Scales: From the Nearby Universe to the First Galaxies**, STScI, Baltimore, MD
- Jun. 20-24, 2022 “The Impact of Stellar Feedback on the Dynamics and Evolution of Star Forming Regions,” Poster, **From Stars to Galaxies II**, Chalmers University, Gothenberg, Sweden
- Jun. 1-3, 2020 “Investigating the Impact of Stellar Feedback on Models of Star Formation,” iPoster, The **236th AAS Meeting**, Online
- Jan. 6, 2017 “From the Ultraviolet to the Infrared: The Stars of M70,” Poster, **229th AAS Meeting**, Grapevine, TX
- Jan 6, 2016 “Star Formation in Nearby Analogues of Lyman Break Galaxies,” Poster, **227th AAS Meeting**, Kissimmee, FL

MEMBERSHIPS AND ACTIVITIES (SELECTED)

- 2015 - Present **American Astronomical Society (AAS)**
Junior Member Fall 2015 to Fall 2017 *and* Feb. 2020 to Dec. 2024
Full Member Dec. 2024 to Present
- 2015 - 2018 **American Physical Society (APS)**, Member Fall 2015 to Fall 2018

OTHER MEETINGS, CONFERENCES, AND WORKSHOPS (SELECTED)

- Sept. 9-20, 2024 **KITP: Cosmic Origins: The First Billion Years**, Program Participant, UC Santa Barbara, Santa Barbara, CA
- Apr. 22-26, 2024 **Advanced AMUSE Workshop**, AMNH, New York, NY
- Jan. 6, 2024 **Peer Review Workshop**, 243rd AAS Meeting, New Orleans, LA
- Apr. 5, 2023 **Torch Regional Meeting**, AMNH, New York, NY
- Jan. 17, 2020 **New England Star Formation Meeting**, University of Connecticut, Storrs, CT

Aug. 28-30, 2019 **Torch Open Source Workshop: Introduction to Structure and Use**, CCA, Flatiron Institute, New York, NY
Jul. 15-26, 2019 **Advancing Theoretical Astrophysics** Summer School, University of Amsterdam, Amsterdam, The Netherlands

OTHER EXPERIENCE

Jul. 10, 2017 to Technical Consultant, Kolisch Hartwell, P.C., Portland, OR
May 31, 2018

PUBLICATIONS

† - indicates an undergraduate student mentee

THESIS

Appel, S. M., “*Exploring the effect of protostellar feedback on star formation and molecular cloud dynamics*”, 2024, Rutgers University, [doi:10.7282/t3-p3kg-dk21](https://doi.org/10.7282/t3-p3kg-dk21)

REFEREED PUBLICATIONS

(Submitted & Under Review) Assilkhan, A., **Appel, S. M.**, Shukirgaliyev, B., Abdikamalov, E., Portegies Zwart, S., Andersson, E. P., Kalambay, M., & Mac Low, M.-M., “*Centrally concentrated star formation in young clusters II: Jet feedback*”, 2026, arXiv e-prints, arXiv:2606.13666, (doi: [10.48550/arXiv.2606.13666](https://doi.org/10.48550/arXiv.2606.13666)).

→ Member of the **Torch** collaboration; contributed to manuscript writing and to discussion of project development and analysis.

Appel, S. M., Burkhardt, B., Mac Low, M.-M., Andersson, E. P., Cournoyer-Cloutier, C., Lewis, S., McMillan, S. L. W., Polak, B., Portegies Zwart, S., Tran, A., & Wilhelm, M. J. C., “*Protostellar Jets in Star Cluster Formation and Evolution: I. Implementation and Initial Results*”, 2026, AJ, 172, 1, (doi: [10.3847/1538-3881/ae633f](https://doi.org/10.3847/1538-3881/ae633f)).

→ Primary author; led the code development, analysis, writing, and submission.

Cournoyer-Cloutier, C., Andersson, E. P., **Appel, S. M.**, Lahén, N., Polak, B., Rantala, A., Toonen, S., Sills, A., Rieder, S., Portegies Zwart, S., Mac Low, M.-M., & Harris, W. E., “*Massive Interacting Binaries Enhance Feedback in Star-forming Regions*”, 2025, ApJ, 990, 112, (doi: [10.3847/1538-4357/ade01](https://doi.org/10.3847/1538-4357/ade01)).

→ Member of the **Torch** collaboration; contributed discussion, comments, and edits.

Polak, B., Mac Low, M.-M., Klessen, R. S., Portegies Zwart, S., Andersson, E. P., **Appel, S. M.**, Cournoyer-Cloutier, C., Glover, S. C. O., & McMillan, S. L. W., “*Massive star cluster formation: III. Early mass segregation during cluster assembly*”, 2025, A&A, 695, A188, (doi: [10.1051/0004-6361/202451785](https://doi.org/10.1051/0004-6361/202451785)).

→ Member of the **Torch** collaboration; contributed discussion, comments, and edits.

†Kiihne, A., **Appel, S. M.**, Burkhardt, B., Semenov, V. A., & Federrath, C., “*Fitting Methods for Probability Distribution Functions in Turbulent Star-forming Clouds*”, 2025, ApJ, 979, 89, (doi: [10.3847/1538-4357/ad99d5](https://doi.org/10.3847/1538-4357/ad99d5)).

→ Guided mentee during project development, analysis, writing, and peer review.

Cournoyer-Cloutier, C., Sills, A., Harris, W. E., Polak, B., Rieder, S., Andersson, E. P., **Appel, S. M.**, Mac Low, M.-M., McMillan, S., & Portegies Zwart, S., “*Massive Star Cluster Formation with Binaries. I. Evolution of Binary Populations*”, 2024, ApJ, 977, 203, (doi: [10.3847/1538-4357/ad90b3](https://doi.org/10.3847/1538-4357/ad90b3)).

→ Member of the **Torch** collaboration; contributed discussion, comments, and edits.

Polak, B., Mac Low, M.-M., Klessen, R. S., Portegies Zwart, S., Andersson, E. P., **Appel, S. M.**, Cournoyer-Cloutier, C., Glover, S. C. O., & McMillan, S. L. W., “*Massive star cluster formation: II. Runaway stars as fossils of subcluster mergers*”, 2024, A&A, 690, A207, (doi: [10.1051/0004-6361/202450774](https://doi.org/10.1051/0004-6361/202450774)).

→ Member of the **Torch** collaboration; contributed discussion, comments, and edits.

Polak, B., Mac Low, M.-M., Klessen, R. S., Wei Teh, J., Cournoyer-Cloutier, C., Andersson, E. P., **Appel, S. M.**, Tran, A., Lewis, S. C., Wilhelm, M. J. C., Portegies Zwart, S., Glover, S. C. O., Rieder, S., Wang, L., & McMillan, S. L. W., “*Massive star cluster formation: I. High star formation efficiency while resolving feedback of individual stars*”, 2024, A&A, 690, A94, (doi: [10.1051/0004-6361/202348840](https://doi.org/10.1051/0004-6361/202348840)).

→ Member of the **Torch** collaboration; contributed discussion, comments, and edits.

Appel, S. M., Burkhardt, B., Semenov, V. A., Federrath, C., Rosen, A. L., & Tan, J. C., “*What Sets the Star Formation Rate of Molecular Clouds? The Density Distribution as a Fingerprint of Compression and Expansion Rates*”, 2023, ApJ, 954, 93, (doi: [10.3847/1538-4357/ace897](https://doi.org/10.3847/1538-4357/ace897)).

→ Primary author; performed most of the analysis and did most of the writing.

Cournoyer-Cloutier, C., Sills, A., Harris, W. E., **Appel, S. M.**, Lewis, S. C., Polak, B., Tran, A., Wilhelm, M. J. C., Mac Low, M.-M., McMillan, S. L. W., & Portegies Zwart, S., “*Early evolution and three-dimensional structure of embedded star clusters*”, 2023, MNRAS, 521, 1338-1352, (doi: [10.1093/mnras/stad568](https://doi.org/10.1093/mnras/stad568)).

→ Member of the **Torch** collaboration; contributed discussion, comments, and edits.

Appel, S. M., Burkhardt, B., Semenov, V. A., Federrath, C., & Rosen, A. L., “*The Effects of Magnetic Fields and Outflow Feedback on the Shape and Evolution of the Density Probability Distribution Function in Turbulent Star-forming Clouds*”, 2022, ApJ, 927, 75, (doi: [10.3847/1538-4357/ac4be3](https://doi.org/10.3847/1538-4357/ac4be3)).

→ Primary author; performed most of the analysis and did most of the writing.

Burkhardt, B., **Appel, S. M.**, Bialy, S., Cho, J., Christensen, A. J., Collins, D., Federrath, C., Fielding, D. B., Finkbeiner, D., Hill, A. S., Ibáñez-Mejía, J. C., Krumholz, M. R., Lazarian, A., Li, M., Mocz, P., Mac Low, M.-M., Naiman, J., Portillo, S. K. N., Shane, B., Slepian, Z., & Yuan, Y., “*The Catalogue for Astrophysical Turbulence Simulations (CATS)*”, 2020, ApJ, 905, 14, (doi: [10.3847/1538-4357/abc484](https://doi.org/10.3847/1538-4357/abc484)).

→ Contributing numerical simulations from **Torch**.

OTHER ABSTRACTS (SELECTED)

†Javier, F., Appel, S., & Mac Low, M.-M., “*Evolution of Gas Properties Due to Protostellar Jets in Star-Forming Clouds*”, 2026, AAS, 247, 409.05.

→ Guided mentee during project and presentation development.

Appel, S. M., Andersson, E. P., Burkhardt, B., Cournoyer-Cloutier, C., Lewis, S., Mac Low,

M.-M., McMillan, S. L. W., Polak, B., Portegies Zwart, S., Tran, A., & Wilhelm, M. J. C., “*Exploring the Role of Protostellar Jets in Simulations of Star Cluster Formation using Torch*”, 2025, AAS, 245, 343.08.

→ Lead author and presenter.

Appel, S., Burkhart, B., Semenov, V., Federrath, C., Rosen, A., & Tan, J., “*The Density Distribution and the Gas Compression and Expansion Rates of Molecular Clouds: Understanding the Star Formation Rate and Exploring the Effect of Protostellar Feedback*”, 2024, AAS, 243, 431.02D.

→ Lead author and presenter; dissertation talk.

Appel, S. M., Burkhart, B., & Federrath, C., “*Investigating the Impact of Stellar Feedback on Models of Star Formation*”, 2020, AAS, 236, 236.03.

→ Lead author and presenter.

Appel, S., Zurek, D., & Leigh, N., “*From the Ultraviolet to the Infrared: The Stellar Population of the Globular Cluster M70*”, 2017, AAS, 229, 343.13.

→ Lead author and presenter as an REU student.

Appel, S., Baker, A. J., & Hall, K., “*Spatially resolved star-formation in nearby analogues of Lyman break galaxies*”, 2016, AAS, 227, 234.08.

→ Lead author and presenter as an REU student.